RP2354B with in-package 2 MB flash; no external flash IC.

SCP-55 dedicated 5 V adapter; 16 ICs. Card removal clears one latch; firmware sequences voltage and RESET.

USB and power

ファイル: scp55-usb-power.kicad_sch

RP2354B

ファイル: scp55-controller.kicad_sch

Card interface

ファイル: scp55-card-interface.kicad_sch

Card power and interlock

ファイル: scp55-card-control.kicad_sch

Card detect

ファイル: scp55-card-detect.kicad_sch

MIDI IN

ファイル: scp55-midi-in.kicad_sch

I2S DAC

ファイル: scp55-dac.kicad_sch

Measurement points

ファイル: scp55-debug.kicad_sch

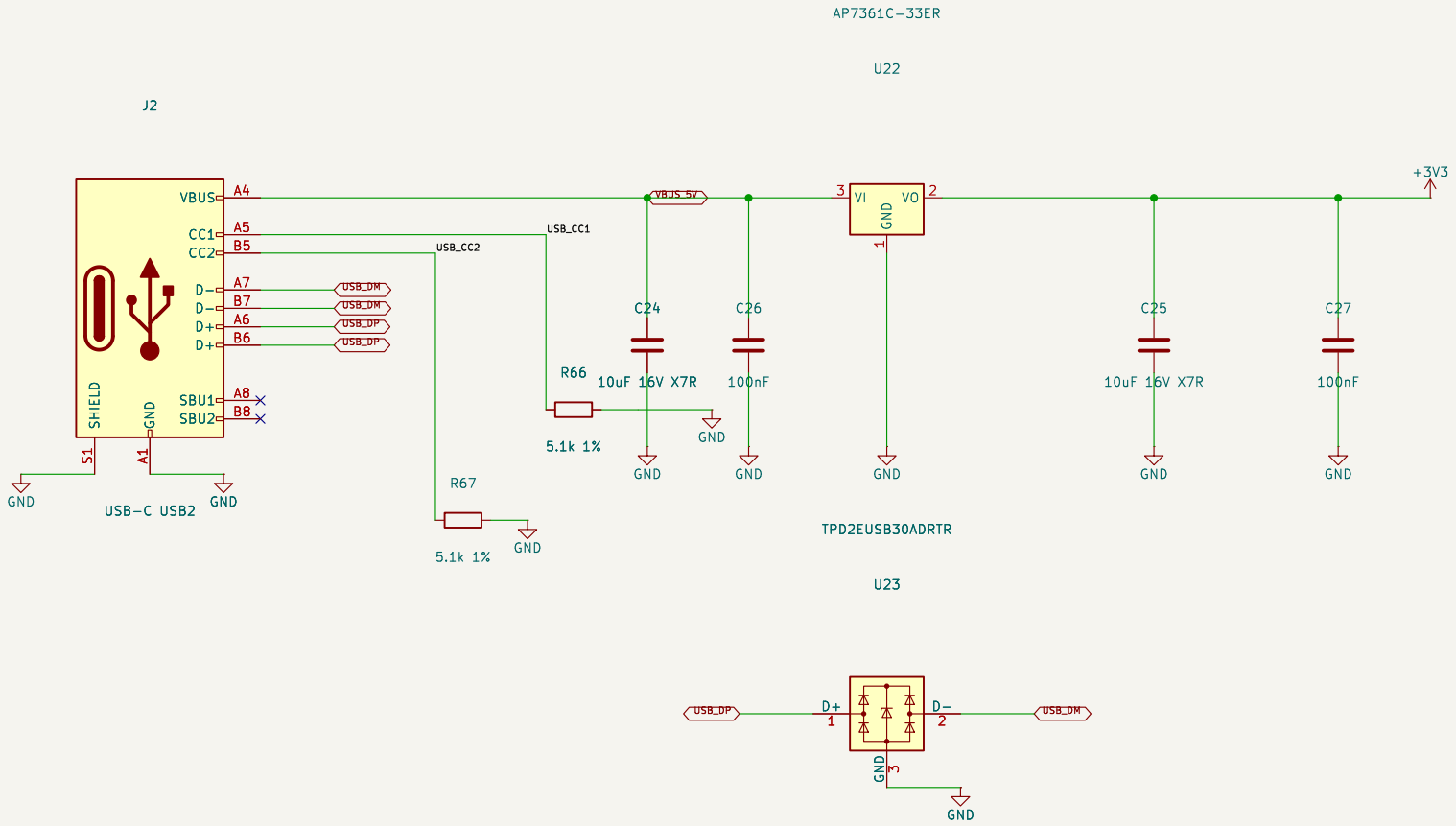
Sheet: /
File: scp55-reva.kicad_sch

**Title: SCP-55 Rev.A Integrated Schematic**

| | | |
|---|---|---|
| Size: A3 | Date: | **Rev: LOGICAL** |
| KiCad E.D.A. 9.0.6 | | Id: 1/9 |

USB-C 5V sink / USB2 device. Physical receptacle footprint to be matched to selected part.

VBUS_5V, 3V3, USB_DP and USB_DM join the bus / RP2354B circuits by named nets.




| | | |
|---|---|---|
| Sheet: /USB and power/ | | |
| File: scp55-usb-power.kicad_sch | | |
| **Title: SCP-55 Rev.A USB-C and 3V3 Input** | | |
| Size: A4 | Date: | **Rev: LOGICAL** |
| KiCad E.D.A. 9.0.6 | | Id: 2/9 |

RP2354B QFN80 + exposed pad; 2 MB in-package flash. Same pinout as RP2350B.

QSPI data/clock are connected internally. Do not attach an external flash; keep external stubs short.

## Boot and 12 MHz clock

R71
1k
SW1
BOOTSEL
X1
ABM8-272-T3
C28
15pF C0G 5%
C29
15pF C0G 5%
Core supply
R74
1k
C33
100nF
R73
33R
L1
3.3uH
C30
4.7uF
C31
4.7uF
C32
4.7uF

U24
RP2354B

## USB and local bypass

R68
27R
R69
27R

C34 100nF
C35 4.7uF
C36 100nF
C37 100nF
C38 100nF
C39 100nF
C40 100nF
C41 100nF
C42 100nF
C43 100nF
C44 100nF
C45 100nF
C46 100nF
C47 100nF

## Reset and SWD

R72
10k
J3
SWD

Sheet: /RP2354B/
File: scp55-controller.kicad_sch

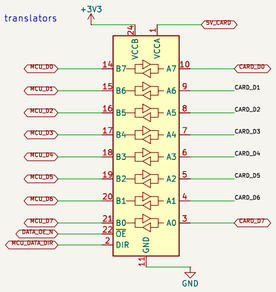
**Title: SCP-55 Rev.A RP2354B Controller**

Size: A3 | Date: | Rev: LOGICAL
KiCad E.D.A. 9.0.6 | Id: 3/9

Socket J1, five level translators, card-side defaults and local bypass capacitors.

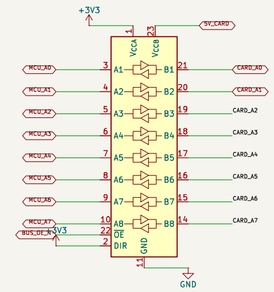
U1

Bidirectional level translators



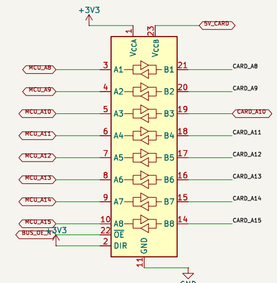

74LVC4245APW

U2




SN74LVC8T245PW

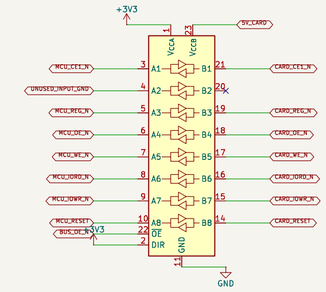
U3




Signal pull-downs

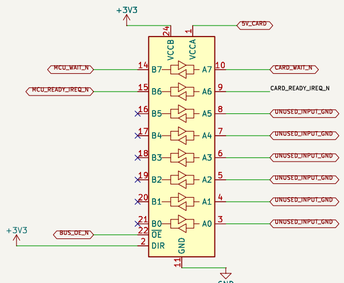
SN74LVC8T245PW

U4




SN74LVC8T245PW

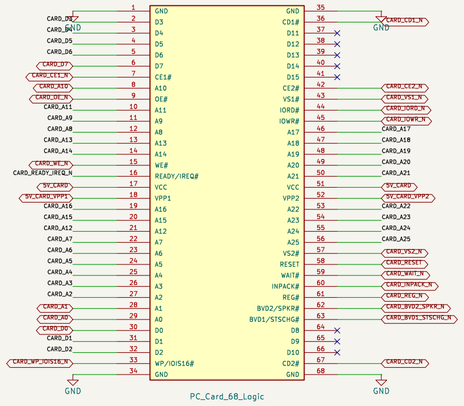
U6



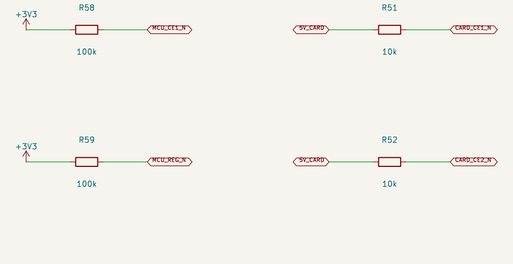

74LVC4245APW

PC Card socket

J1



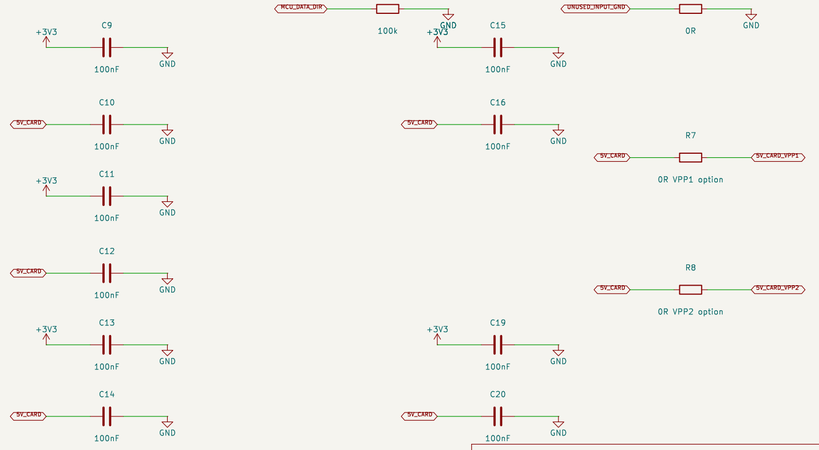

Card-side defaults

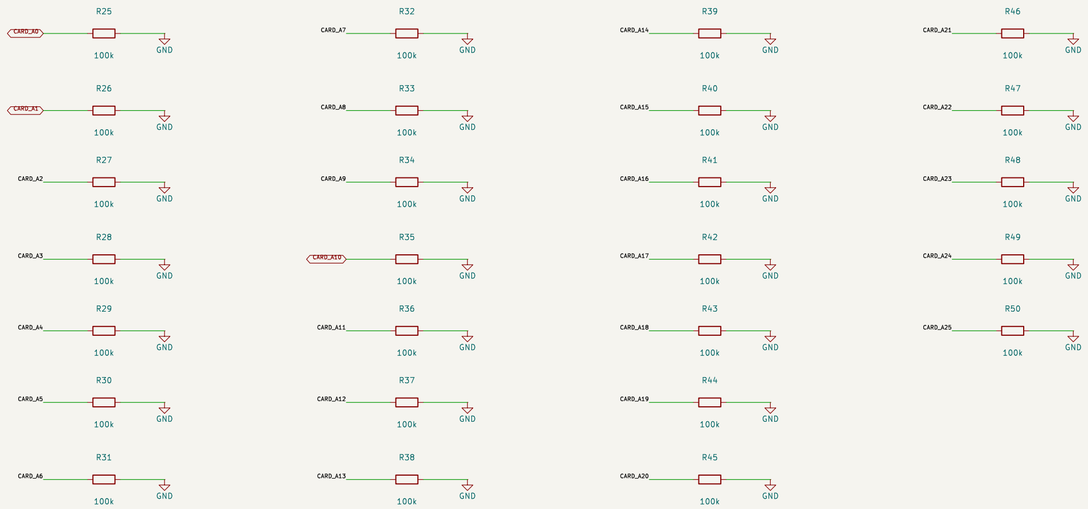






Local bypass and options



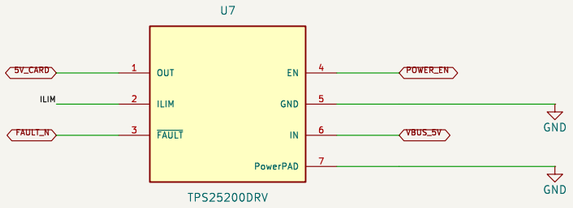

Sheet: /Card interface/
File: scp55-card-interface.kicad_sch

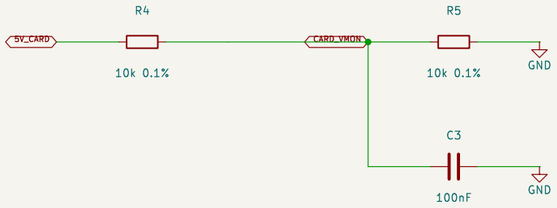
**Title: SCP-55 Rev.A: Card socket and bus translators**

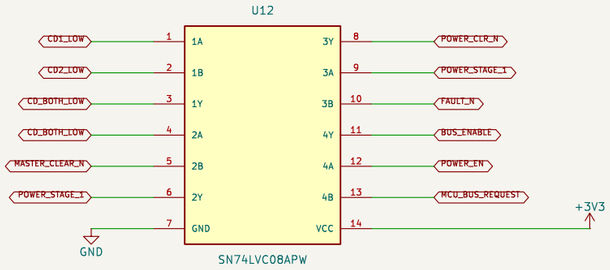
| Size: A2 | Date: | Rev: LOGICAL |
|---|---|---|
| KiCad E.D.A. 9.0.6 | | Id: 4/9 |

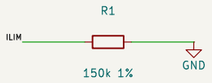
SCP-55 dedicated 5 V adapter: single removal latch; no automatic restart.

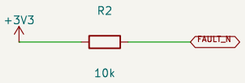
Firmware checks ADC voltage and settling time before enabling the bus and applying RESET.

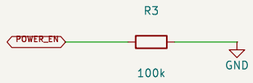


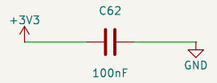



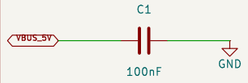



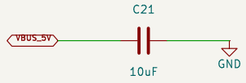



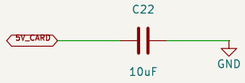



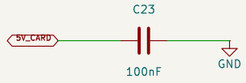



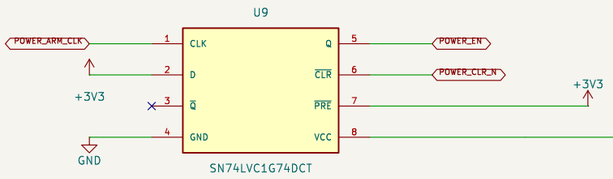



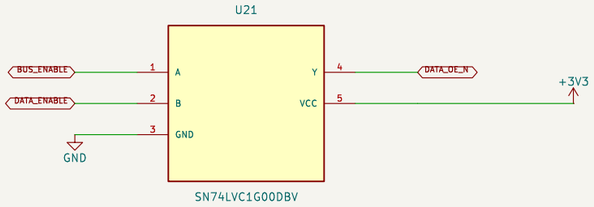



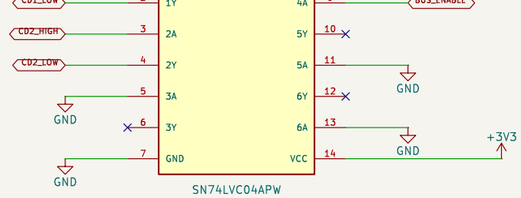



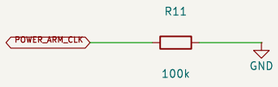



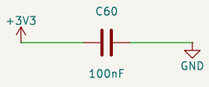



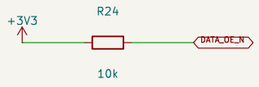



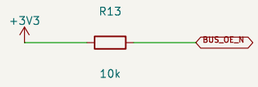



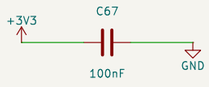



















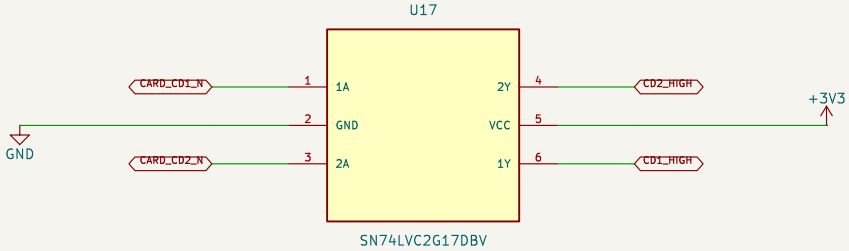
Sheet: /Card power and interlock/
File: scp55-card-control.kicad_sch

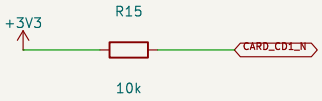
**Title: SCP-55 Rev.A: Card power and interlock**

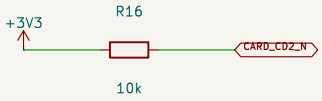
| Size: A3 | Date: | Rev: LOGICAL |
|---|---|---|
| KiCad E.D.A. 9.0.6 | | Id: 5/9 |

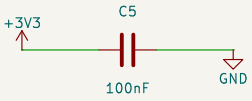
SCP-55 only. CD1/CD2 open -> asynchronous power and bus shutdown.

10k pull-ups to 3.3 V. VS1/VS2 remain on socket test points only.



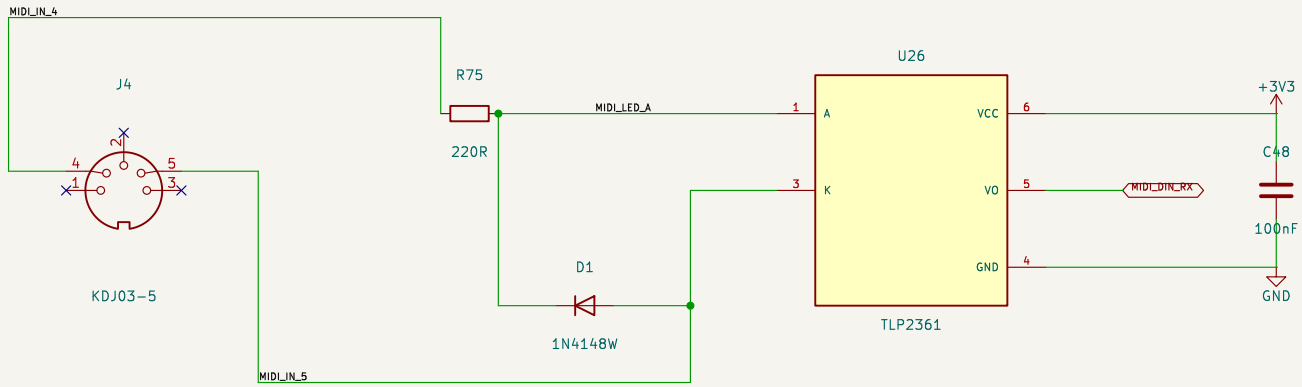




| | | |
|---|---|---|
| Sheet: /Card detect/ | | |
| File: scp55-card-detect.kicad_sch | | |
| **Title: SCP-55 Rev.A: Card detect (SCP-55 only)** | | |
| Size: A4 | Date: | **Rev: LOGICAL** |
| KiCad E.D.A. 9.0.6 | | Id: 6/9 |

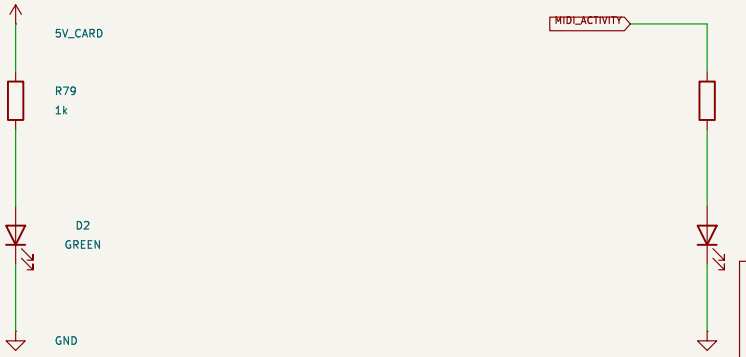
DIN pin 2 / shield has no DC path to circuit GND; pins 1 and 3 are unused.

TLP2361 output is inverted: LED current gives UART low. Keep C48 within 1 cm of pins 6 and 4.




PWR / green / card 5 V supply

MIDI / yellow / 100 ms activity hold



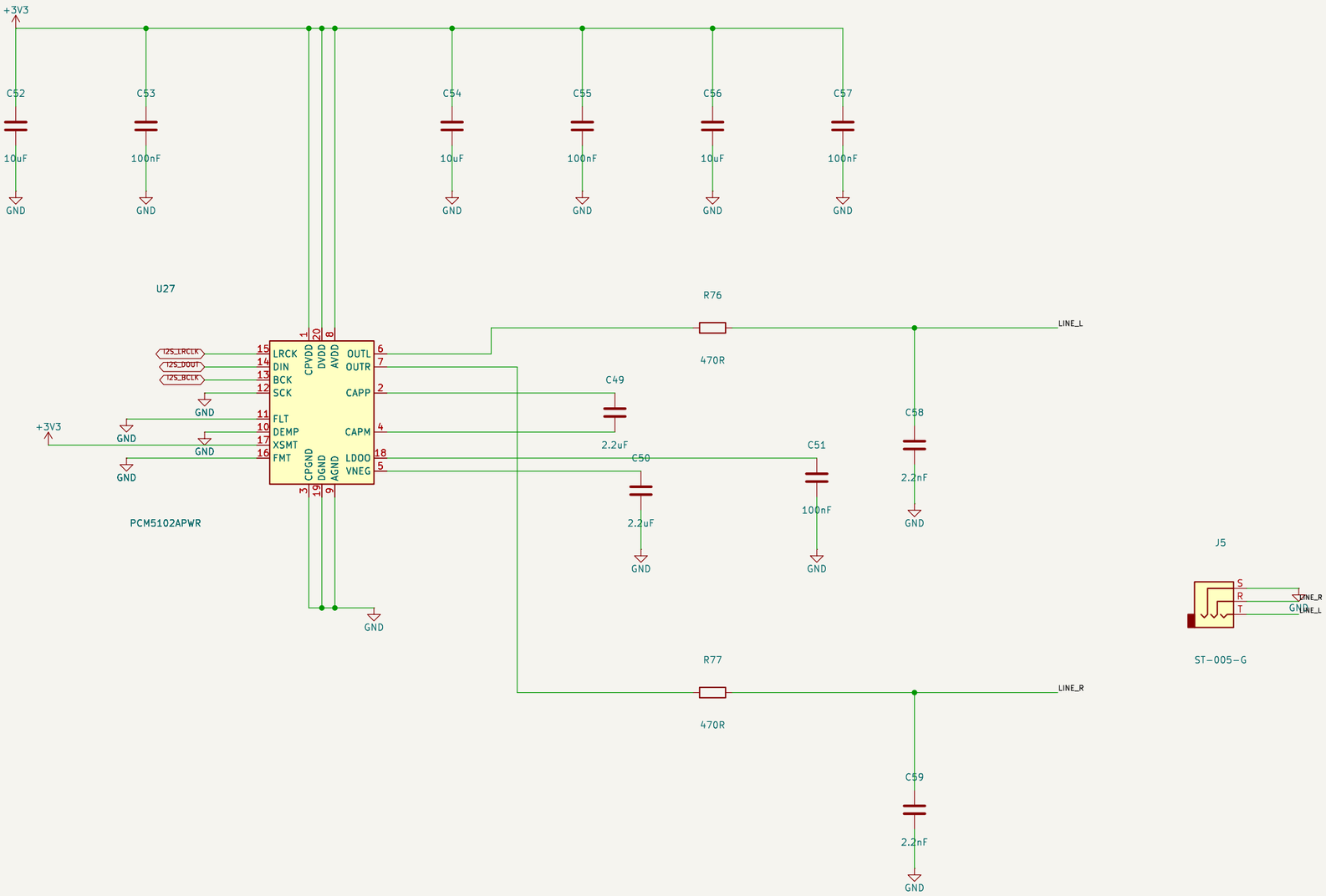

Sheet: /MIDI IN/
File: scp55-midi-in.kicad_sch

**Title: SCP-55 Rev.A MIDI IN**

| Size: A4 | Date: | **Rev: LOGICAL** |
|---|---|---|
| KiCad E.D.A. 9.0.6 | | Id: 7/9 |

PCM5102A 3-wire I2S: SCK=GND, FMT=DEMP=FLT=GND; line output only.

470R + 2.2nF per output per TI Figure 33. Place charge-pump and supply capacitors at DAC pins.




| | | |
|---|---|---|
| Sheet: /I2S DAC/ | | |
| File: scp55-dac.kicad_sch | | |
| **Title: SCP-55 Rev.A I2S DAC and LINE OUT** | | |
| Size: A3 | Date: | **Rev: LOGICAL** |
| KiCad E.D.A. 9.0.6 | | Id: 8/9 |

PWR_FLAG symbols declare the external USB and switched card rails to KiCad ERC; they are not physical parts.

VBUS_5V
TP1
5V_CARD
TP2
+3V3
TP3
GND
TP4
5V_CARD_VPP1
TP5
5V_CARD_VPP2
TP6
CARD_RESET
TP7
CARD_REG_N
TP8
CARD_CE1_N
TP9
CARD_CE2_N
TP10
CARD_IORD_N
TP11
CARD_IOWR_N
TP12
CARD_OE_N
TP13
CARD_WE_N
TP14
CARD_WAIT_N
TP15
CARD_A0
TP16
CARD_A1
TP17
CARD_A10
TP18
CARD_D0
TP19
CARD_D7
TP20
BUS_OE_N
TP21
DATA_OE_N
TP22
MCU_DATA_DIR
TP23
POWER_EN
TP24
CARD_VMON
TP25
BUS_ENABLE
TP26
CARD_WP_IOIS16_N
TP27
CARD_INPACK_N
TP28
CARD_BVD2_SPKR_N
TP29
CARD_BVD1_STSCHG_N
TP30
CARD_CD1_N
TP31
CARD_CD2_N
TP32
CARD_VS1_N
TP33
CARD_VS2_N
TP34
VBUS_5V
#FLG1
PWR_FLAG
5V_CARD
#FLG2
PWR_FLAG
GND
#FLG3
PWR_FLAG

Sheet: /Measurement points/
File: scp55-debug.kicad_sch

**Title: SCP-55 Rev.A Measurement Points**

Size: A3 | Date: | **Rev: LOGICAL**
KiCad E.D.A. 9.0.6 | Id: 9/9